

Technical Report

Calculating the Length of an Event Horizon

1. Executive Summary

The event horizon of a black hole marks the boundary beyond which no signal can escape. Its “length” is most naturally expressed as the circumferential radius (or circumference) of the horizon in a coordinate system that is regular at the horizon. For a non-rotating, uncharged black hole this reduces to the well-known Schwarzschild radius, while for rotating (Kerr) or charged (Reissner–Nordström) black holes the horizon radius depends on spin and charge. This report consolidates the key theoretical concepts, outlines a practical computational workflow, and presents a serverless-style Kubernetes architecture that can be used to calculate the horizon circumference on demand.

2. Research Findings

Category	Key Points
Important Concepts	• Event Horizon – null surface beyond which nothing can escape a black hole. • Schwarzschild Radius – ($r_s = \frac{2GM}{c^2}$) for a non-rotating, uncharged black hole. • Kerr / Reissner–Nordström Metrics – generalizations for rotating or charged black holes. • Proper Length / Circumference – spatial distance measured along a spacelike slice; for a spherical horizon the circumference ($C = 2\pi r_s$). • Coordinate Systems – Schwarzschild, Kerr–Schild, Painlevé–Gullstrand, etc., each giving different coordinate “lengths” due to coordinate singularities. • Geodesic Integration – numerical integration of null geodesics to trace the horizon. • Numerical Relativity – solving Einstein’s equations on a grid to locate the apparent horizon.
Requirements	1. Mass (M) of the black hole (and charge Q, spin a if applicable). 2. Choice of metric (Schwarzschild, Kerr, etc.) appropriate to the physical situation. 3. Coordinate system that is regular at the horizon (e.g., Kerr–Schild). 4. Mathematical tools: differential geometry, tensor calculus, and, for rotating holes, solving the horizon equation ($g^{\mu\nu}\partial_\mu r \partial_\nu r = 0$). 5. Computational resources for numerical integration or horizon-finding algorithms (e.g., AHFinderDirect, Quasi-Local Horizon Finder). 6. Observational data (if the goal is to compare with astrophysical measurements).

Benefits	• Fundamental Insight – Determines the size of the “point of no return,” crucial for understanding black hole thermodynamics and Hawking radiation. • Astrophysical Modeling – Accurate horizon size is needed for accretion disk simulations, jet launching models, and gravitational-wave waveform generation. • Testing General Relativity – Comparing theoretical horizon sizes with observations (e.g., Event Horizon Telescope images) tests the theory in the strong-field regime. • Educational Value – Provides a concrete example of how geometry and physics intertwine in GR.
Challenges	• Coordinate Dependence – The “length” can vary with the slicing of spacetime; one must distinguish between proper length, coordinate radius, and circumferential radius. • Rotating Black Holes – The horizon is oblate; its shape depends on spin, making analytic expressions more complex. • Numerical Instabilities – Horizon-finding algorithms can be sensitive to grid resolution and gauge choices. • Observational Limitations – Direct measurement of horizon size is indirect; uncertainties in mass, distance, and inclination propagate into horizon estimates. • Quantum Corrections – At Planck scales, classical GR may break down, altering the notion of a horizon.

Quick Reference: How to Compute the “Length” (Circumference) of a Schwarzschild Horizon

1. **Determine the mass (M)** of the black hole (from dynamics, lensing, or gravitational waves).
2. Compute the Schwarzschild radius: